

What AI Agents Are and How They Work

Chatbot vs agent · How an agent decides · Giving it tools · Memory · When one agent isn't enough

From chatbot to agent

A chatbot	An agent
Answers your question	Works towards a goal
One question, one answer	Many steps, on its own
Only knows what it's told	Can look things up and take action
You do the work	It does the work, you check it



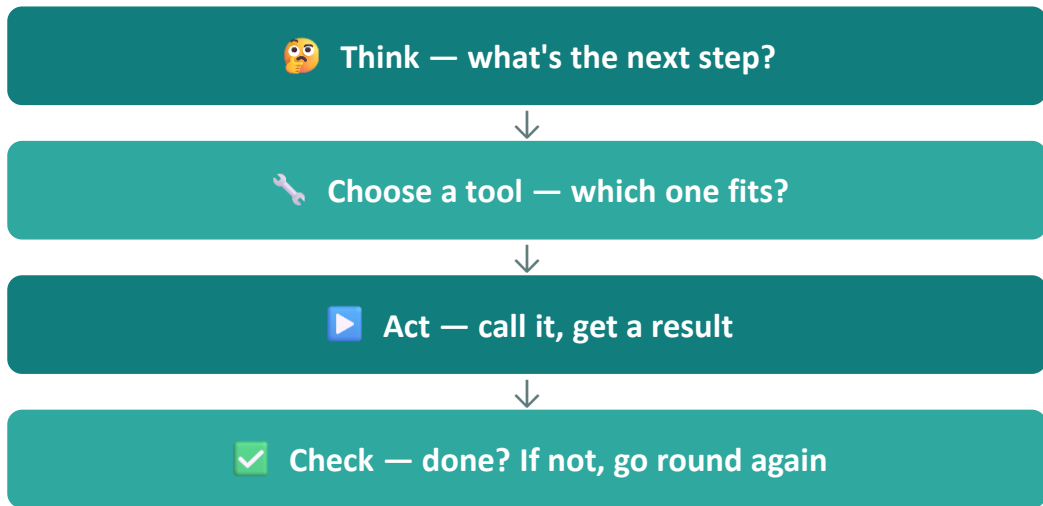
"What's our refund policy?" → a chatbot answers.



"Refund this customer's order and email them." → that needs an agent.

How an agent actually works

The agent runs in a loop until the job is done:



The model isn't doing the work. It's deciding what to do next — and something else does each job.

Giving an agent tools

An agent can only do what you let it do. You hand it a list of tools.

How you describe a tool	What it is
Action groups	The classic Bedrock Agents way
OpenAPI schema	Point it at an API you already have
MCP	A shared standard so any agent can use any tool



A tool is just a function with a plain-English description.

The model reads that description to decide when to use it — so a clear description isn't documentation, it's what makes the agent choose correctly.

Memory

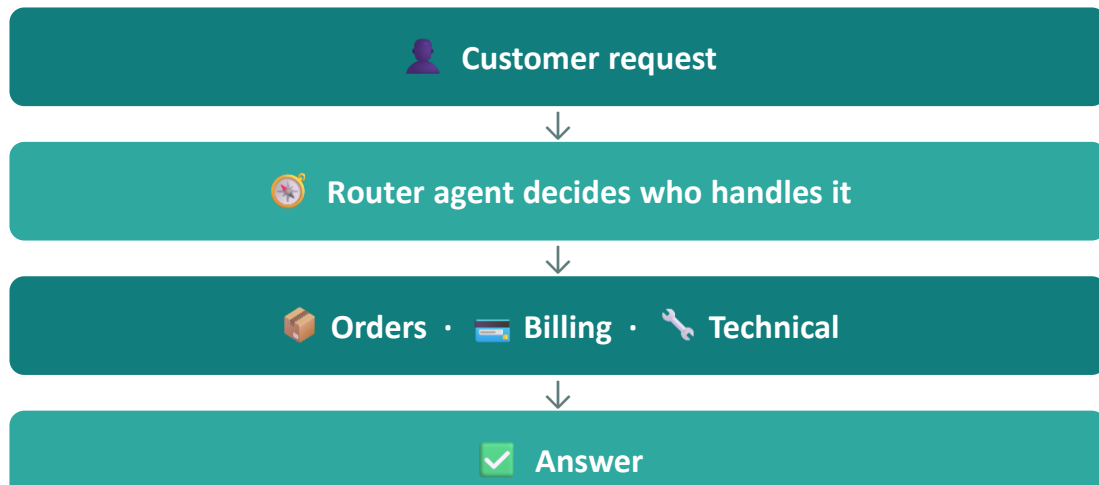
Short-term memory	Long-term memory
This conversation	Across conversations
"the order I just mentioned"	"she always wants express delivery"
Gone when you close the chat	Stored and looked up later

 **Long-term memory means you're storing what people told you.**

That's a privacy decision, not just a technical one — what you keep, how long, and whether someone can ask you to delete it.

When one agent isn't enough

Give one agent thirty tools and it starts picking the wrong one. So you split the job up.



💡 Same idea as a support team: a front desk that routes, and specialists who go deep. Each agent stays small — and when something goes wrong, you can tell which one got it wrong.

Humans in the loop

Some steps shouldn't happen without a person.

The step	Who decides
Look up an order	✔ Let the agent do it
Draft a reply	✔ Let the agent do it
Refund a large amount	⏸ Ask a human first
Delete a record	⏸ Ask a human first



The useful question isn't "can the agent do this?" It's "**what happens if it does it wrong?**"

Key takeaways

- 1 A chatbot answers. An agent does.
- 2 It runs a loop: think → pick a tool → act → check
- 3 It can only do what you give it tools for
- 4 Long-term memory is a privacy decision too
- 5 Ask "what if it's wrong?", not "can it do this?"

Next · Part 2: the AWS services that actually build these